

NETWORK SECURITY & SCANNING

Detailed Study & Practical Report

1. Objective

The objective of this task is to understand reconnaissance, network scanning, vulnerability assessment, network traffic analysis, and basic firewall configuration in an authorized cybersecurity lab environment.

The practical work was performed using **Kali Linux** as the testing machine and **Metasploitable2** as the intentionally vulnerable target.

Target IP: 192.168.56.101

2. Lab Environment

Tools Used

- Kali Linux
- Metasploitable2
- Nmap
- Whois
- Nslookup
- Google Search
- Shodan
- OpenVAS/GVM
- Wireshark
- hping3
- iptables

The testing was performed within the private virtual lab network.

[INSERT SCREENSHOT — Kali + Metasploitable2 lab/network setup, if you have one]

3. Passive Reconnaissance

Passive reconnaissance collects information about a target without directly interacting with its systems.

For the initial reconnaissance exercises, example.com was used as a safe public training target.

3.1 Whois

Purpose

Whois is used to obtain publicly available domain registration information such as registrar information, domain dates, and name servers.

Command

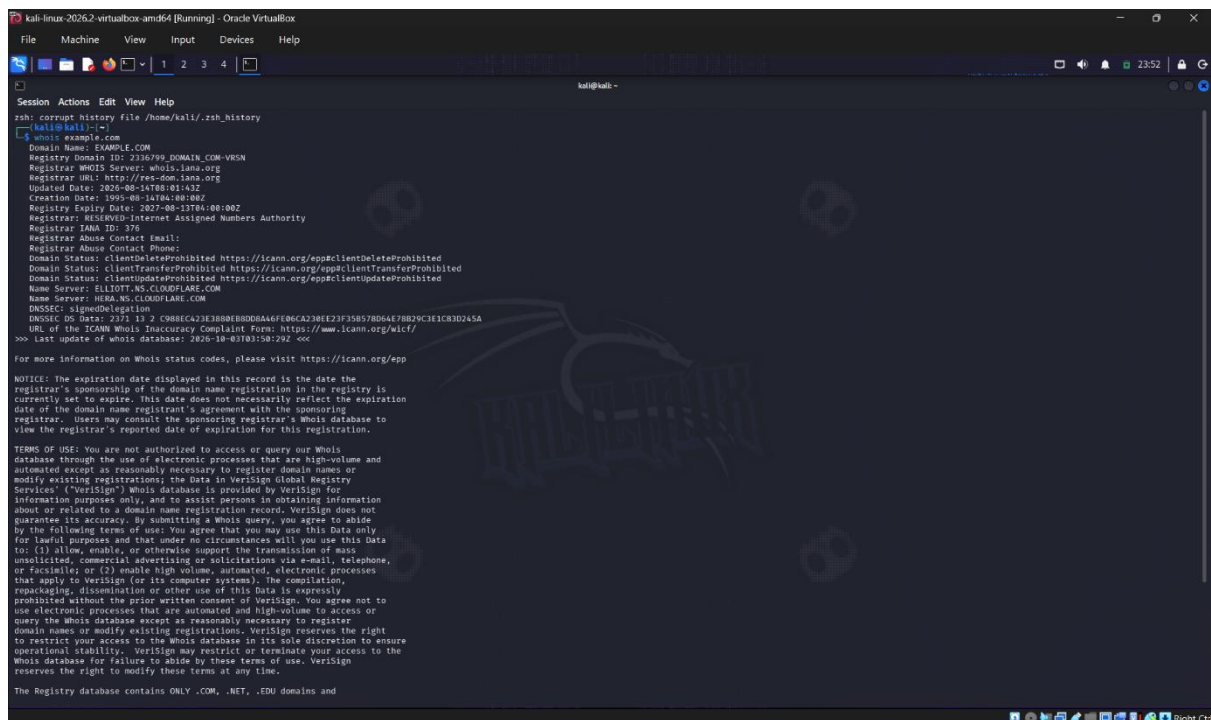
whois example.com

Observation

The command returned publicly available registration/domain information for example.com.

Security Relevance

Attackers can use publicly available registration information as part of information gathering before attempting further reconnaissance.



```
kali-linux-20262-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~$ whois example.com
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://rs-who.iana.org
Updated Date: 2026-08-14T08:01:43Z
Creation Date: 1995-08-14T04:08:00Z
Registry Expiry Date: 2027-08-13T04:08:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: ELLIOTT.NS.CLOUDFLARE.COM
Name Server: HEBA.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 C98EC423E388B8BDD8AA6FE66CA238EE23F35B78D64E78829C3E1C83D245A
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-10-01T01:58:29Z +0000

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
```

3.2 Nslookup

Purpose

Nslookup is a DNS troubleshooting and information-gathering utility. It can be used to resolve domain names into IP addresses and examine DNS records.

Command

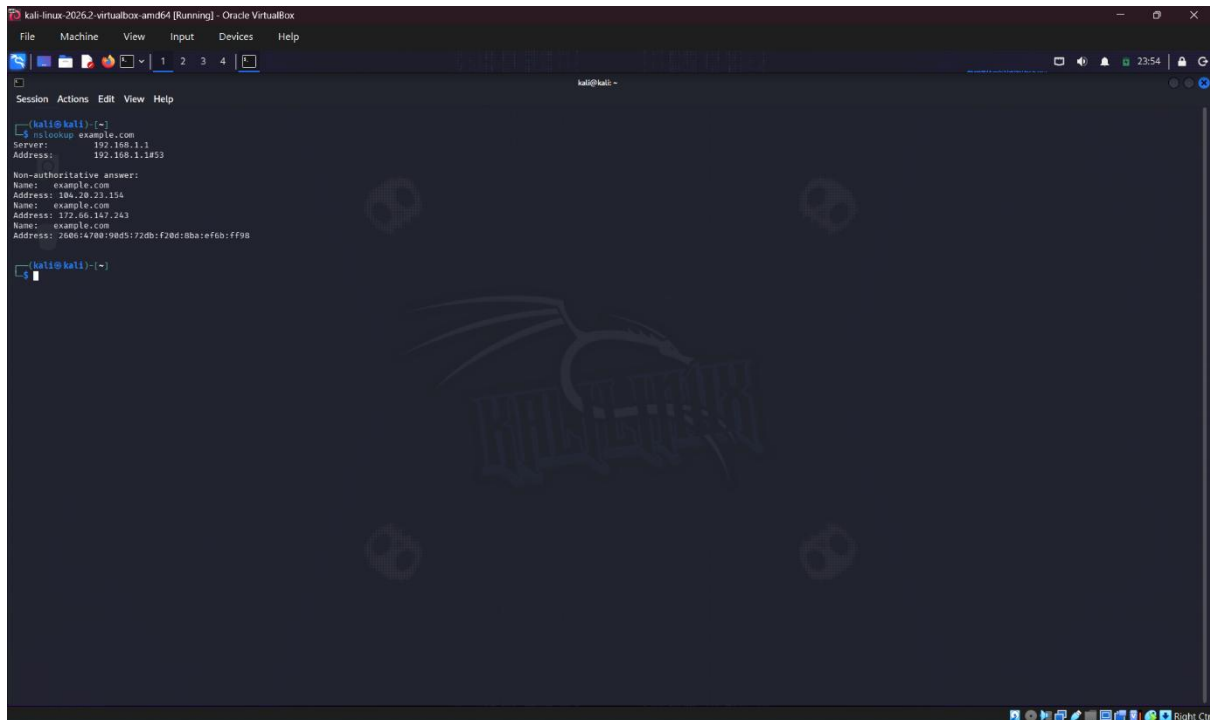
nslookup example.com

Observation

The command resolved the domain and displayed its DNS information.

Security Relevance

DNS information can help identify infrastructure associated with a domain.

A screenshot of a Kali Linux terminal window. The terminal shows the output of the 'nslookup example.com' command. The output includes the server (192.168.1.1), the address (192.168.1.1#53), and a non-authoritative answer with the name (example.com) and three addresses (184.20.23.154, 172.68.147.243, and 2606:14788:98d5:72db:f20d:bba:ef0b:ff98). The terminal window has a dark background with a 'Kali Linux' watermark.

```
kali@kali:~$ nslookup example.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   example.com
Address: 184.20.23.154
Name:   example.com
Address: 172.68.147.243
Name:   example.com
Address: 2606:14788:98d5:72db:f20d:bba:ef0b:ff98

kali@kali:~$
```

3.3 Google Dorking

Purpose

Google Dorking uses advanced search operators to locate publicly indexed information.

Searches Used

site:example.com

and

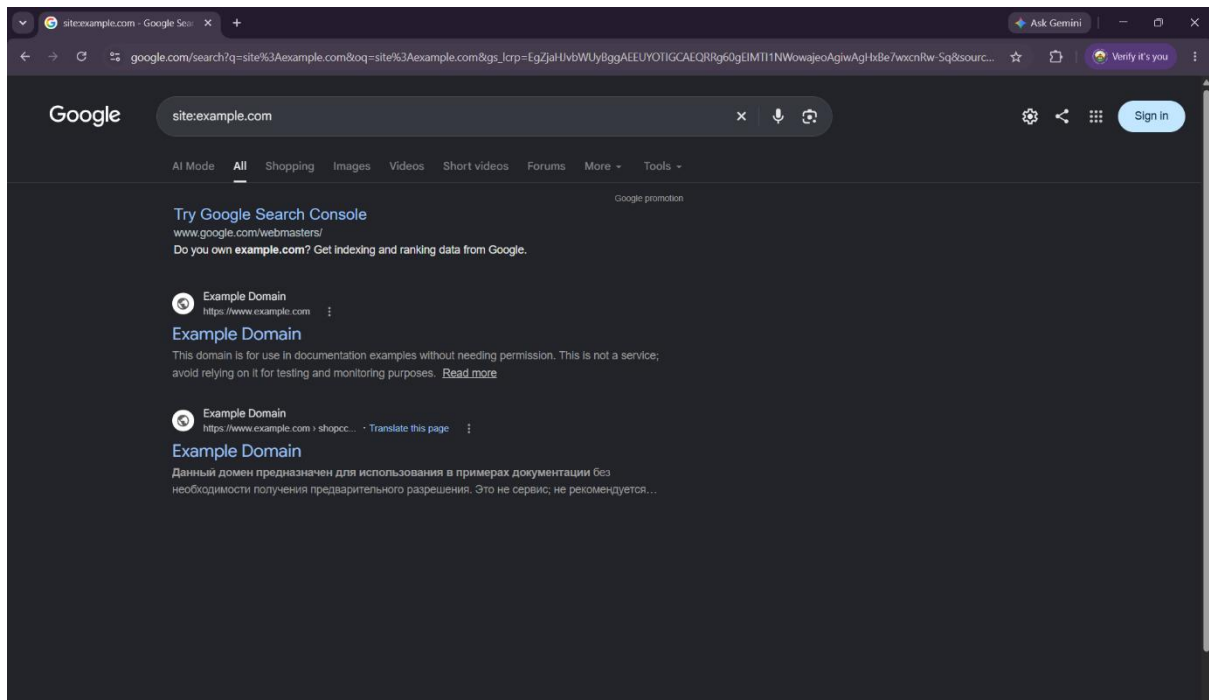
site:example.com filetype:pdf

Observation

The searches demonstrated how search operators can restrict results to a particular domain or file type.

Security Relevance

Organizations may unintentionally expose documents or information through publicly indexed content.



3.4 Shodan

Purpose

Shodan is a search engine that indexes information about internet-connected services and devices.

Practical

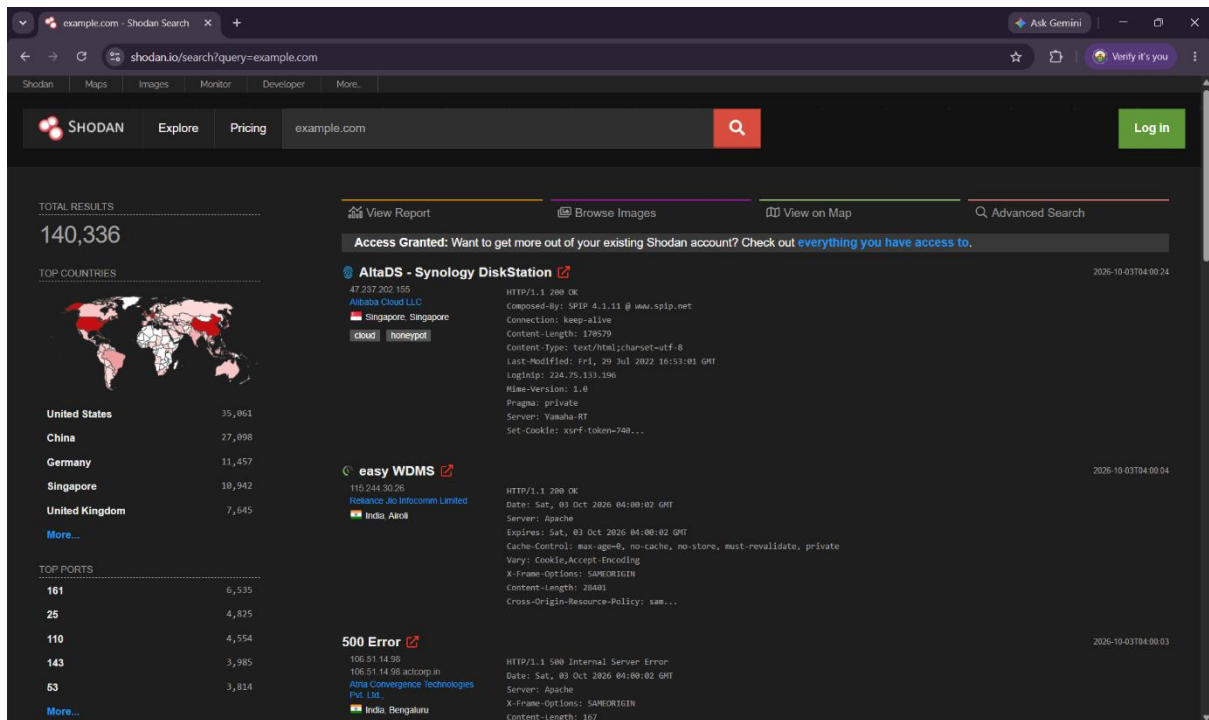
The domain example.com was searched in Shodan.

Observation

No significant useful indexed host information was obtained from the search.

Security Relevance

Shodan can provide security researchers with information about publicly exposed services.



4. Active Reconnaissance

Active reconnaissance directly interacts with systems in the authorized lab to identify hosts and services.

4.1 Ping Sweep / Host Discovery

Purpose

A ping sweep identifies active hosts within a network.

Command

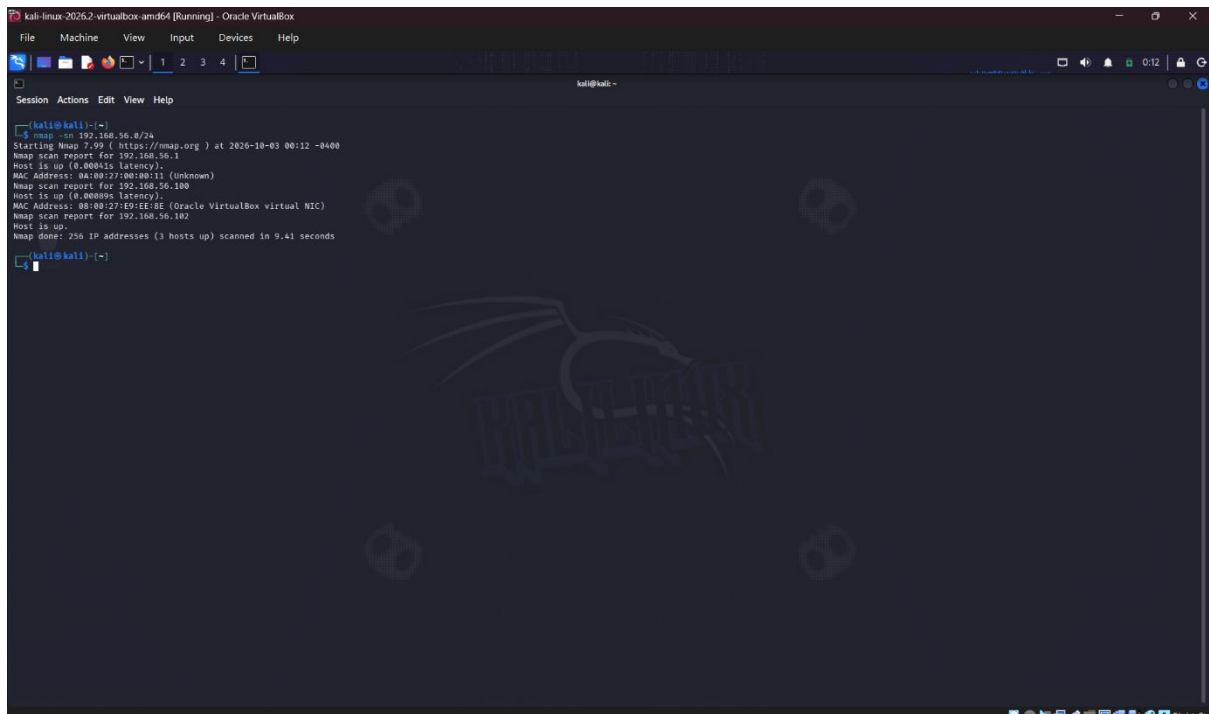
```
nmap -sn 192.168.56.0/24
```

Observation

The scan identified active systems in the private lab network, including the Metasploitable2 target.

Security Relevance

Host discovery helps identify systems that may be investigated during a security assessment.



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali -
Session Actions Edit View Help
kali@kali:~$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2020-10-03 00:12 -9400
Nmap scan report for 192.168.56.1
Host is up (0.00041s latency).
MAC Address: 0A:00:27:00:00:11 (unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00000s latency).
MAC Address: 08:00:27:E9:EE:8E (Oracle VM VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 9.41 seconds
kali@kali:~$
```

4.2 Banner / Service Detection

Purpose

Banner grabbing identifies services and their versions running on a target.

Command

```
nmap -sV 192.168.56.101
```

Observation

Multiple services were identified on Metasploitable2, including FTP, SSH, Telnet, HTTP, DNS, MySQL, PostgreSQL, VNC and other services.

Security Relevance

Service and version information can help security testers identify outdated or potentially vulnerable software.

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali -
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap -sS 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2020-10-03 00:45 -9400
Nmap scan report for 192.168.56.101
Host is up (0.00055s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshexec
513/tcp   open  login        OpenSSH or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1899/tcp  open  java-rmi     GNU Classpath gnuiregistry
1924/tcp  open  bindshell    Metasploitables root shell
2849/tcp  open  nfs          2.4 (RPC #100000)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.6.51a-ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5986/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  x11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8089/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
NAC Address: 00:00:27:71:89:42 (Oracle VM VirtualBox virtual NIC)
Service Info: hosts: metasploitable.localdomain, irc: metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.28 seconds
kali@kali:~$
```

5. Nmap TCP SYN Scan

Purpose

A TCP SYN scan identifies open TCP ports without completing the full TCP connection.

Command

```
nmap -sS 192.168.56.101
```

Actual Result

The scan identified **23 open TCP ports**, including:

Port Service

- 21 FTP
- 22 SSH
- 23 Telnet
- 25 SMTP
- 53 DNS
- 80 HTTP
- 111 RPC
- 139 NetBIOS

Port Service

445 SMB

3306 MySQL

5432 PostgreSQL

5900 VNC

6667 IRC

8009 AJP13

8180 HTTP

Additional open ports were also identified by the scan.

Security Observation

The target exposed a large number of network services. In a real environment, unnecessary exposed services increase the attack surface.

6. Nmap UDP Scan

Purpose

UDP scanning identifies services operating over UDP.

Command

```
nmap -sU 192.168.56.101
```

Actual Result

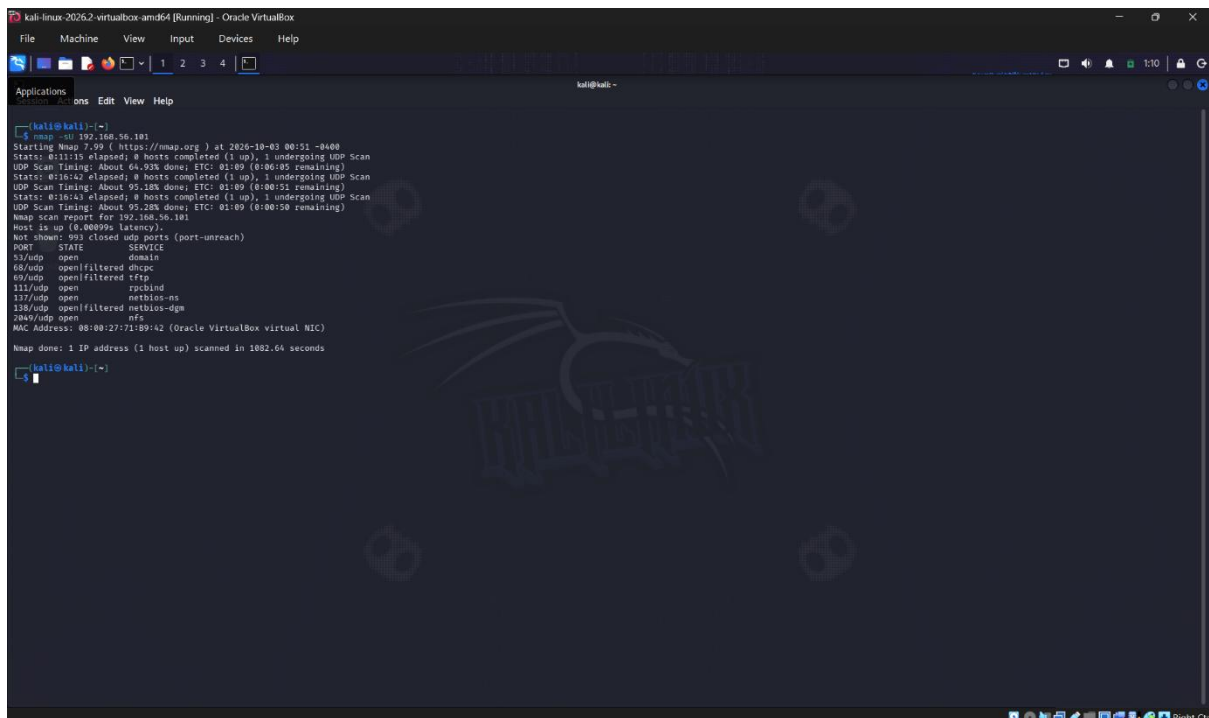
The scan identified UDP services including:

- UDP 53 — DNS
- UDP 68 — DHCP
- UDP 69 — TFTP
- UDP 111 — RPC
- UDP 137 — NetBIOS
- UDP 138 — NetBIOS
- UDP 2049 — NFS

Some ports were reported as open|filtered.

Security Observation

UDP services must also be considered during network security assessments because TCP-only scanning does not identify them.



```
kali@kali:~$ nmap -sU 192.168.56.101
Starting Nmap 7.90 ( https://nmap.org ) at 2020-10-03 00:51 -0400
State: 0:11:15 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 64.92% done; ETC: 01:09 (0:00:05 remaining)
State: 0:15:43 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 95.18% done; ETC: 01:09 (0:00:51 remaining)
State: 0:16:43 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 95.28% done; ETC: 01:09 (0:00:50 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0000ms latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE    SERVICE
53/udp    open     domain
68/udp    openFiltered dhcp
69/udp    openFiltered rftp
111/udp    open     rpcbind
137/udp    open     netbios-ns
138/udp    openFiltered netbios-dgm
2049/udp   open     nfs
MAC Address: 08:00:27:71:89:A2 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1002.64 seconds
kali@kali:~$
```

7. Service Version Detection

Purpose

Service version detection identifies the software and version associated with discovered ports.

Command

```
nmap -sV 192.168.56.101
```

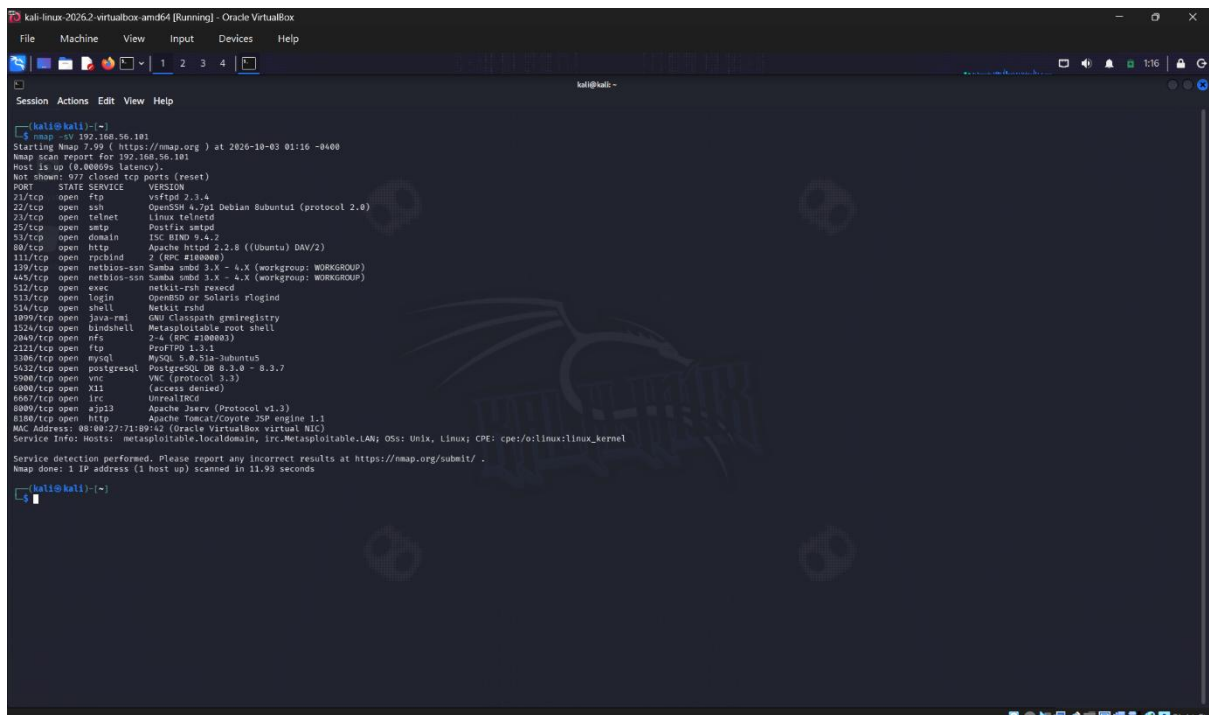
Important Results

Examples from the actual scan included:

- FTP — vsftpd 2.3.4
- SSH — OpenSSH 4.7p1
- DNS — ISC BIND 9.4.2
- HTTP — Apache 2.2.8
- MySQL — 5.0.51a
- PostgreSQL — 8.3.x
- IRC — UnrealIRCd
- Tomcat — Apache Tomcat/Coyote

Security Observation

Version information is useful for vulnerability assessment because outdated software versions may have known security issues.



```
kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.90 ( https://nmap.org ) at 2026-10-03 01:10 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0000ms latency).
Not shown: 577 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache/2.2.8 ((Ubuntu) DAV/2)
113/tcp   open  rsh            2 (RHC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath gmicregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2.4 (RFC 18000)
2121/tcp  open  ftp            ProFTPD 1.3.1
2386/tcp  open  mysql          MySQL 5.0.51a-ubuntu
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6080/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8089/tcp  open  ajp13          Apache Jserv (Protocol 1.3)
8189/tcp  open  http           Apache/2.2.8 (Ubuntu)
MAC Address: 08:00:27:71:89:42 (Oracle VM VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.93 seconds
```

8. Operating System Detection

Purpose

Nmap OS detection attempts to determine the operating system based on network behavior and TCP/IP characteristics.

Command

```
nmap -O 192.168.56.101
```

Actual Result

Nmap identified:

Running: Linux 2.6.X

OS details: Linux 2.6.9 - 2.6.33

Network Distance: 1 hop

Security Relevance

Knowing the operating system helps security testers understand the target environment and determine which security controls or vulnerabilities may be relevant.

```
kali@kali:~$ nmap -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2020-10-03 01:17 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00001s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1899/tcp  open  rmiregistry
1323/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:71:89:42 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.23 seconds
kali@kali:~$
```

9. Vulnerability Scanning — OpenVAS/GVM

Purpose

OpenVAS/GVM is used to perform vulnerability assessments and identify potential security weaknesses.

The Metasploitable2 virtual machine was configured as the intended scan target.

Target

Metasploitable2

192.168.56.101

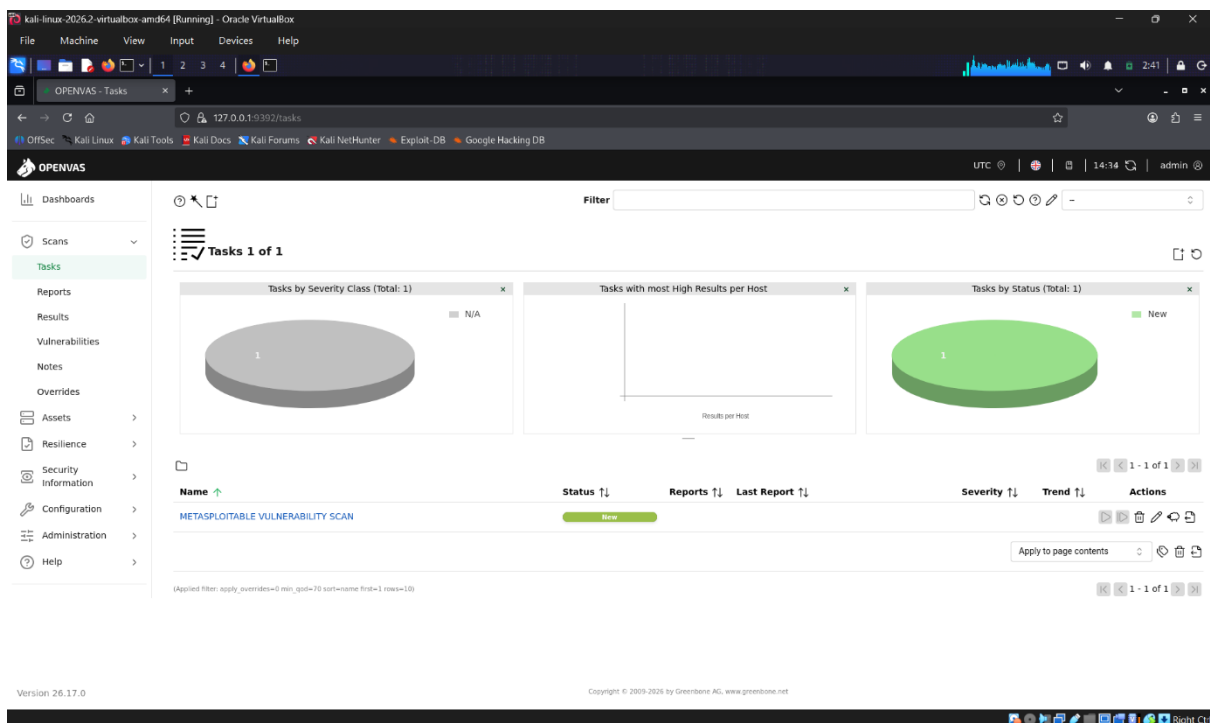
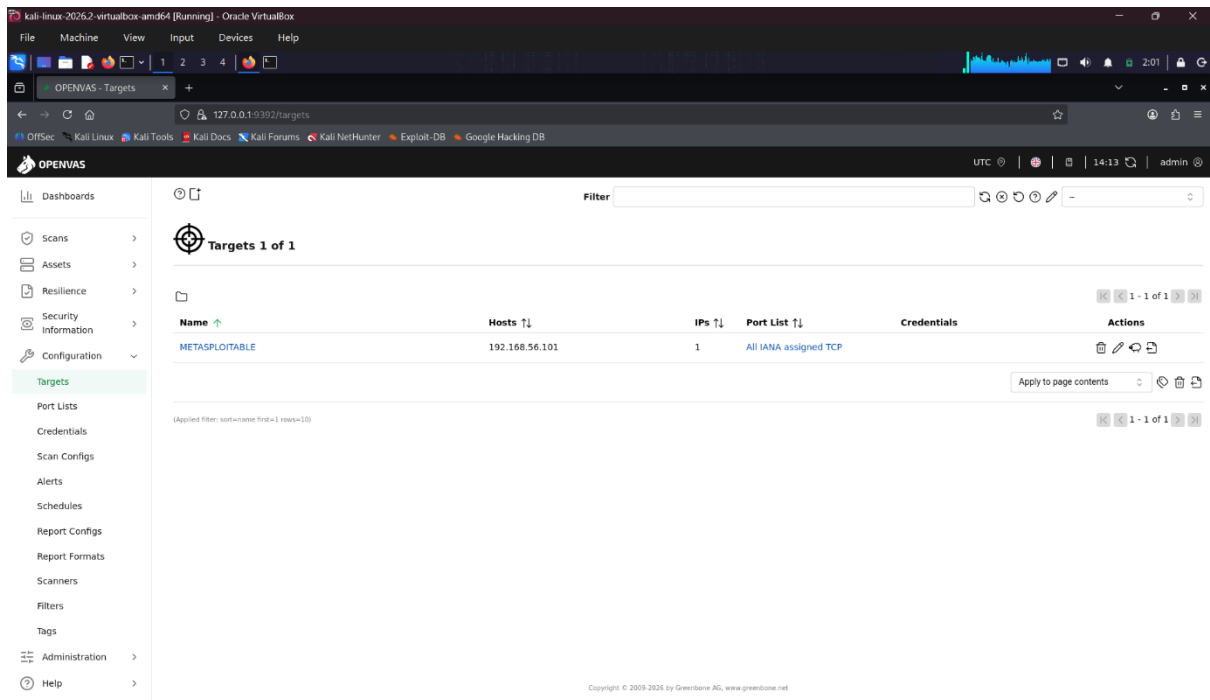
Current Status

OpenVAS/GVM was installed and configured. Feed synchronization was initiated, but the vulnerability scan has **not yet been completed**.

Therefore, **no Critical/High/Medium/Low vulnerability results are being claimed in this report**.

Important

Do not insert a fake vulnerability-results screenshot here.



10. HTTP Traffic Analysis with Wireshark

Purpose

Wireshark was used to capture and analyze HTTP network traffic.

Filter

http

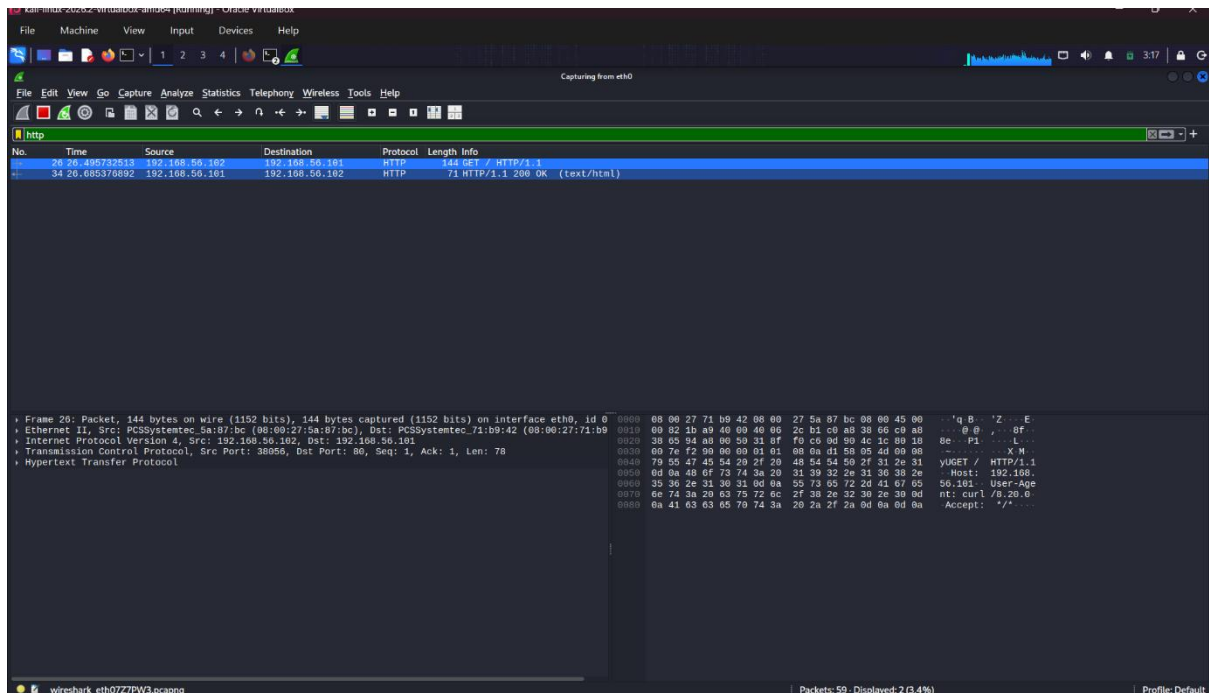
Observation

HTTP requests and responses were observed between Kali Linux and Metasploitable2.

The capture included an HTTP GET request and an HTTP 200 OK response.

Security Observation

HTTP traffic is unencrypted. Information transmitted through HTTP can potentially be observed by someone capable of capturing the traffic.



11. FTP Traffic and Unencrypted Credentials

Purpose

The practical demonstrated why unencrypted FTP traffic is insecure.

Lab Connection

ftp 192.168.56.101

Authorized Metasploitable2 lab credentials were used.

Wireshark Filter

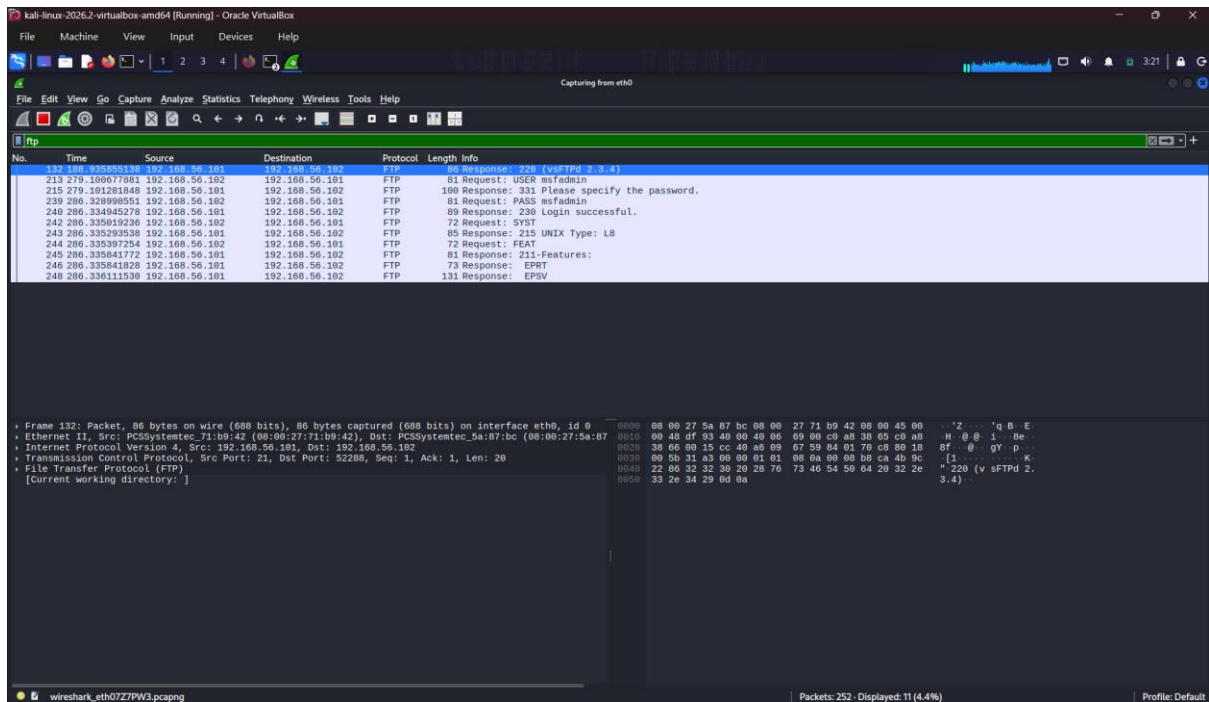
ftp

Observation

FTP authentication traffic was visible in the packet capture, demonstrating that FTP credentials can be transmitted without encryption.

Security Observation

Plain FTP should not be used for sensitive authentication because captured network traffic can expose credentials.



12. DNS Traffic Analysis

Purpose

Wireshark was used to observe DNS request and response traffic.

Wireshark Filter

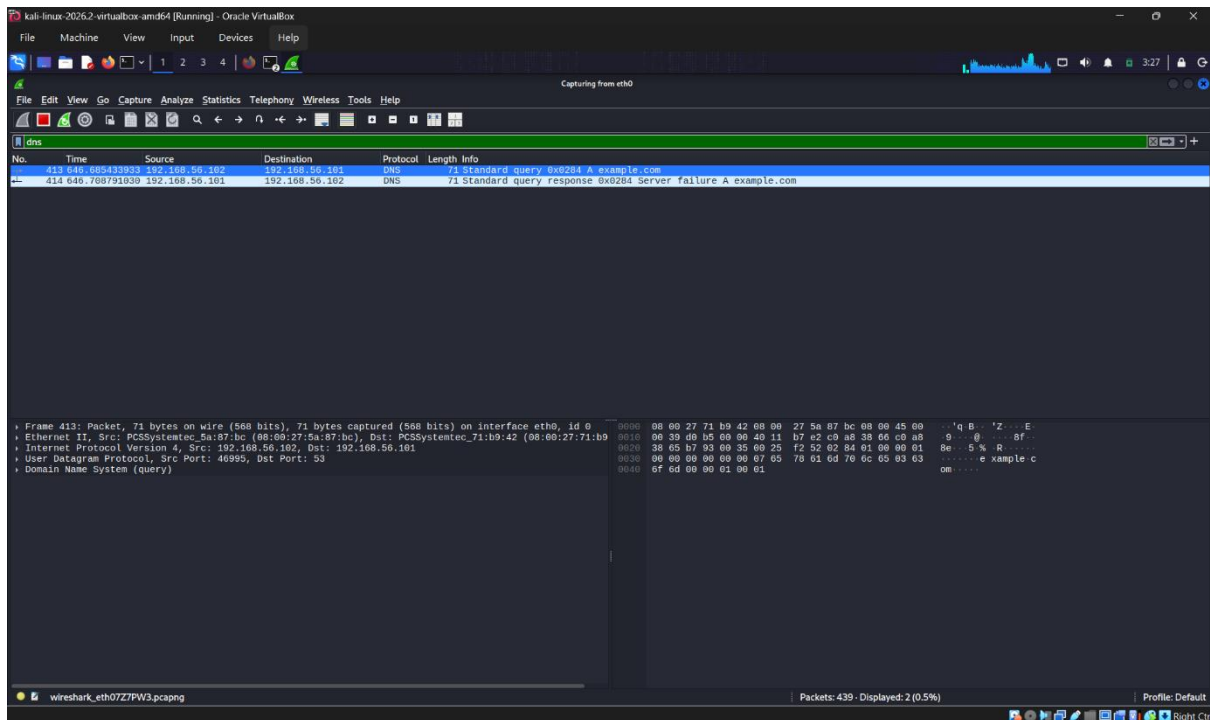
dns

Observation

DNS query and response packets were captured and analyzed.

Security Relevance

DNS traffic provides information about domain-resolution activity occurring on the network.



13. SYN Flood Traffic Analysis

Purpose

The practical demonstrated SYN traffic using a controlled simulation against the authorized Metasploitable2 lab machine.

Command Used

```
sudo hping3 -S -p 80 -c 100 192.168.56.101
```

The -c 100 option bounded the simulation to 100 packets.

Wireshark Filter

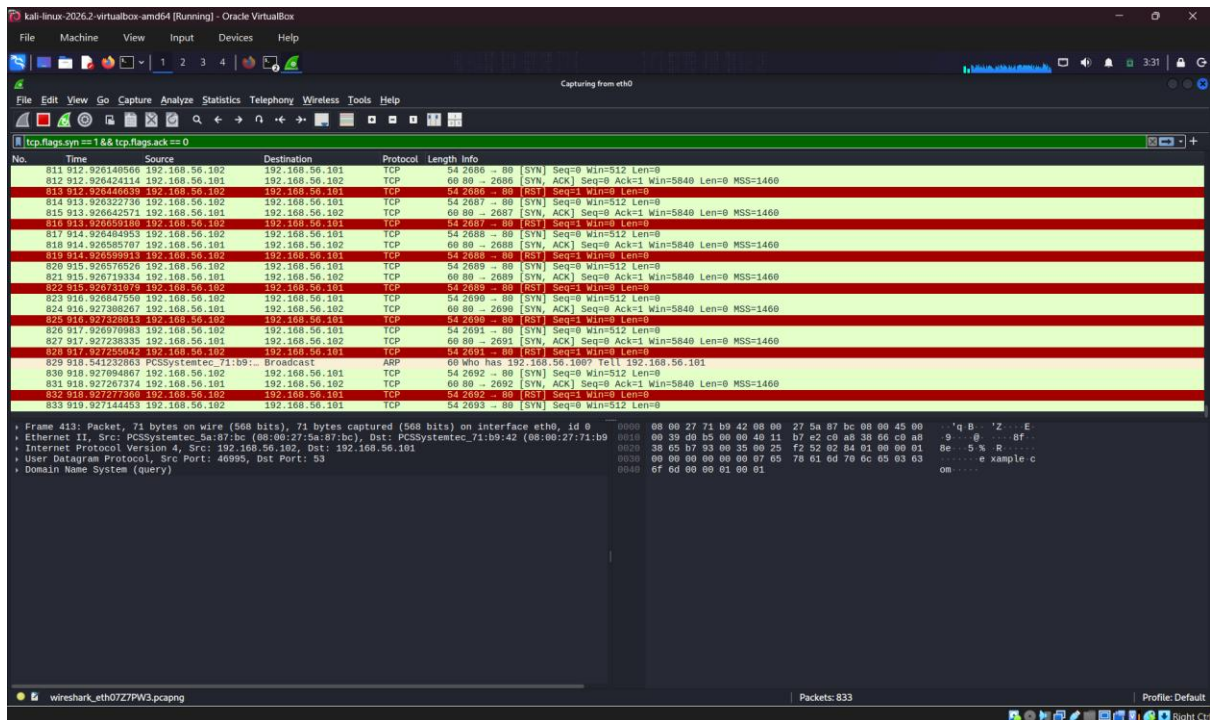
```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Observation

SYN packets targeting TCP port 80 were visible in Wireshark.

Security Concept

A SYN flood attempts to consume resources by generating large numbers of TCP connection requests.



14. Firewall Basics — iptables

Purpose

iptables can be used to control network traffic by creating rules that allow or deny packets.

View Existing Rules

```
sudo iptables -L -n
```

The initial configuration showed:

Chain INPUT

policy ACCEPT

Save Existing Rules

```
sudo iptables-save > ~/iptables-before.txt
```

14.1 Block FTP Port 21

A temporary rule was created:

```
sudo iptables -A INPUT -p tcp --dport 21 -j DROP
```

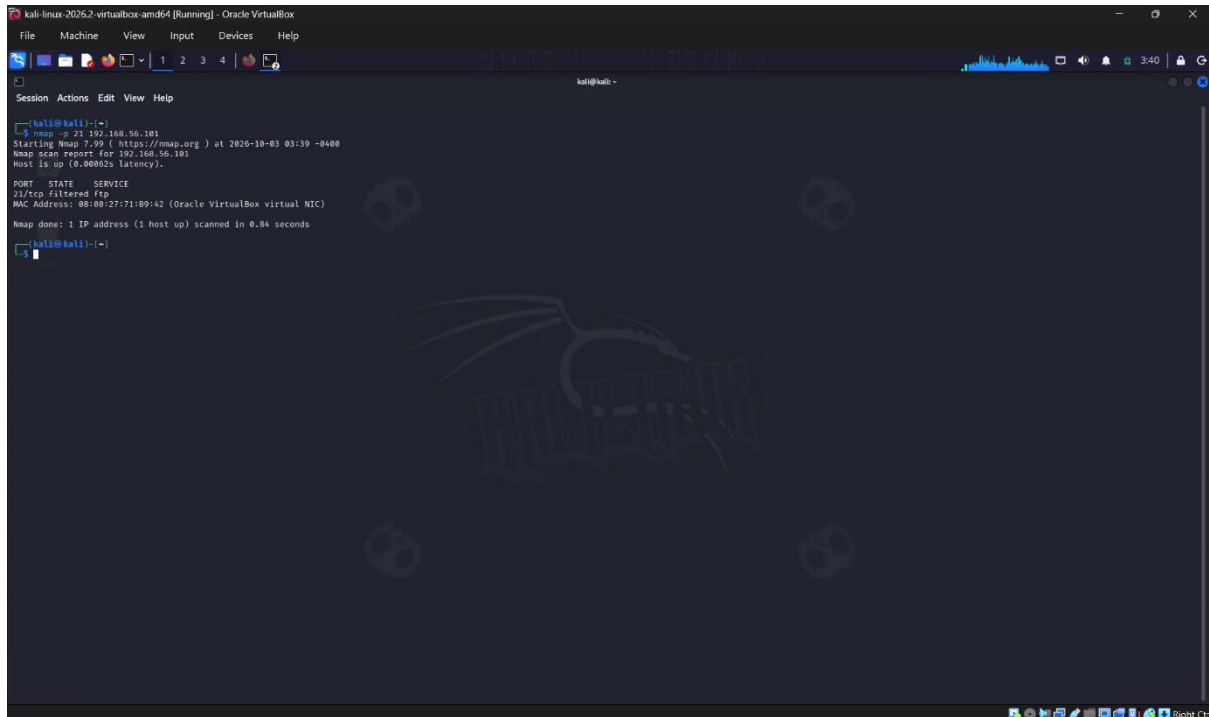
This rule instructs the firewall to drop incoming TCP traffic destined for port 21.

Verification

From Kali:

```
nmap -p 21 192.168.56.101
```

The purpose was to demonstrate that the firewall could prevent access to the selected port.



```
kali@kali:~$ nmap -p 21 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2020-10-03 03:39 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00002s latency).

PORT      STATE SERVICE
21/tcp    filtered ftp
MAC Address: 08:00:27:71:89:42 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
```

14.2 Remove Temporary Rule

After the demonstration, the temporary rule was removed:

```
sudo iptables -D INPUT -p tcp --dport 21 -j DROP
```

This restored the previous firewall state.

15. Key Findings

The practical exercises demonstrated several important security concepts:

Reconnaissance

Publicly available information can provide useful information about domains and infrastructure.

Network Exposure

Metasploitable2 exposed numerous network services, demonstrating how a system with many unnecessary services can have a large attack surface.

Outdated Services

The service detection results showed several old software versions, demonstrating why software inventory and patch management are important.

Unencrypted Protocols

HTTP and FTP demonstrated the risks associated with transmitting information without encryption.

Network Monitoring

Wireshark allowed network packets to be inspected and individual protocols to be analyzed.

Firewall Controls

iptables demonstrated how individual network ports can be blocked and subsequently tested with Nmap.

16. Commands Used

```
whois example.com
```

```
nslookup example.com
```

```
nmap -sn 192.168.56.0/24
```

```
nmap -sS 192.168.56.101
```

```
nmap -sU 192.168.56.101
```

```
nmap -sV 192.168.56.101
```

```
nmap -O 192.168.56.101
```

```
ftp 192.168.56.101
```

```
sudo hping3 -S -p 80 -c 100 192.168.56.101
```

```
sudo iptables -L -n
```

```
sudo iptables-save > ~/iptables-before.txt
```

```
sudo iptables -A INPUT -p tcp --dport 21 -j DROP
```

```
nmap -p 21 192.168.56.101
```

```
sudo iptables -D INPUT -p tcp --dport 21 -j DROP
```

17. Conclusion

Task 2 provided practical experience with reconnaissance, network discovery, port and service scanning, operating-system detection, network traffic analysis, SYN traffic analysis, and basic firewall configuration.

The Nmap assessment demonstrated the services exposed by the Metasploitable2 test machine. Wireshark demonstrated how different protocols appear at packet level, including the security implications of unencrypted HTTP and FTP. The iptables exercise demonstrated how firewall rules can restrict access to individual ports.

The OpenVAS/GVM environment was configured and feed synchronization was initiated. The vulnerability scan itself remains pending, so vulnerability severity results have not been included.